B.S. DEGREE in PHYSICS
Specialized Curriculum in Astronomy and Astrophysics
Iowa State University

Semester 1	Freshman Year	Semester 2	
Critical Thinking and Comm. - Engl 1500	3	**Intro to Class. Physics II – Phys 2420**	5
Introductory Seminar - Phys 1990	R	**Calculus III - Math 2650**	4
Intro to Classical Physics I – Phys 2410	5	**Intermediate Seminar - Phys 2990**	1
Calculus II – Math 1660	4	Natural Science, e.g. General Chemistry I - Chem 1770+1770L	5
Arts&Humanities + IP prespective*	3	Social Science Choice	3
Library Instruction – Lib 1600	1		
	16		18

Semester 1	Sophomore Year	Semester 2	
WOVE Composition - Engl 2500	3	**Intro to Modern Physics II – Phys 3220**	3
Intro to Modern Physics I – Phys 3210	3	**Intro Lab in Mod Phys II – Phys 3220L**	1
Intro Lab in Mod Phys I - Phys 3210L	1	**Classical Mechanics - Phys 3610**	3
Elementary Differential Equations and Laplace Transforms – Math 2670	4	**Introduction to Partial Differential Equations - Math 3850**	3
Stars, Galaxies, and Cosmology - Astro 1500[3]	3	Social Science Choice	3
Humanities choice	3	Humanities Choice	3
	17		16

Semester 1	Junior Year	Semester 2	
Intermediate Mechanics - Phys 3620	3	**Thermal Physics – Phys 3040**	3
Electricity and Magnet. I - Phys 3640	3	**Electricity and Magnet. II – Phys 3650**	3
Introduction to Solar System Astronomy - Astro 3420[3]	3	English Proficiency Requirement - Engl 3020, 3050, 3090 or 3140	3
Matrix/Lin. Algebra – Math 2070/3170	3(4)	Introduction to Astrophysics - Astro 3460[3]	3
Foreign Language (or Elective)	4(3)	Foreign Language (or Elective)	4(3)
	16		16(15)

Semester 1	Senior Year	Semester 2	
Electronics - Phys 3100	4	Intermediate Laboratory – Phys 3110[1]	2
Quantum Mechanics – Phys 4800	3	Elective	3
Elective	3	Applied Quantum Mechanics – Phys 4810[2]	3
Astronomy Laboratory - Astro 3440L[1,3]	3	Humanities Choice	3
Social Science Choice	3	Astrophysical Cosmology - Astro 4050[3]	3
	16		14

* Students should choose a course that satisfy both Arts& Humanities and International Prospective requirements

[1] Students must earn a minimum of two laboratory credits from Phys 3110, 3110T, 4500L, 4700L, Astro 3440L or 4500L.

[2] Recommended but not required. Students must earn 3 credits from Phys 4220, 4810, 4960, 5110, 5260, 5280, 5310, 5340, 5410 or Astro 3420, 3460, 4050.

[3] Recommended for an Astro minor

Students in all ISU majors must complete a three-credit course in U.S. diversity, a three-credit course in international perspectives, 9 credits in Social Science course, 8 credits in Natural Science courses and 12 credits in Arts and Humanities courses.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level. You must also complete the LAS foreign language requirement.